Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/28/2004
PI Name:	Beckermann, Christoph Ph.D.		
Project Title:	Microstructure Evolution in Free Dendritic Growth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1527	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Paley, Mark	**Contact Phone:**	256 544-7775
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNM04AA18G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: (1) to study transient growth of multiple interacting equiaxed dendrites; (2) to study free growth of binary alloy alloy dendrites in the presence of coupled heat and solute transport. The tasks will be accomplished through a combination of experiments, using transparent model substances in an existing dendrite growth setup, and numerical simulations using the phase-field and other methods. The objective of the project is to study microstructural evolution in free dendritic growth of pure substances and alloys from a supercooled melt. The research will not only close important gaps in our current understanding of dendritic growth, but is also needed for placing current simulations of grain structure evolution in castings on firmer theoretical grounds.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The study is important for improving the understanding and computer simulation of solidification processing of metals and alloys.
Task Progress:	During the past year, progress has been made on both tasks of this project. Numerical simulations have been performed to (i) investigate transient dendritic growth of multiple interacting dendrites and (ii) study the dendrite tip operating state selection for free dendritic growth with coupled heat and solute diffusion. The results have been published in journal papers. The experiments involving free dendritic growth of free binary alloy dendrites are still in progress. The experimental and measurement systems have been calibrated and tested for pure succinonitrile. The next step will be to increase the solute concentration of the melt in the growth chamber.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Peer-reviewed Journals	Ramirez JC, Beckermann C, Karma A, Diepers HJ. "Phase-field modeling of binary alloy solidification with coupled heat and solute diffusion." Physical Review E Stat Nonlin Soft Matter Phys. 2004 May;69(5 Pt 1):051607. PMID: 15244829 , May-2004
Articles in Peer-reviewed Journals	Steinbach, I.; Diepers, H.-J.; and Beckermann, C. "Transient Growth and Interaction of Equiaxed Dendrites" J. Crystal Growth (in press) , Jan-2005
Articles in Peer-reviewed Journals	Ramirez, J.C.; Beckermann, C. "Examination of Binary Alloy Free Dendritic Growth Theories with a Phase-Field Model" Acta Materialia (in press) , Jan-2005
Papers from Meeting Proceedings	Ramirez JC, Beckermann C. "Examination of Binary Alloy Free Dendritic Growth Theories with a Phase-Field Model." Solidification Processes and Microstructures - A Symposium in Honor of Wilfried Kurz, eds. M. Rappaz, C. Beckermann, and R. Trivedi, TMS, Warrendale, PA, 2004, pp. 373-378. In: Solidification Processes and Microstructures - A Symposium in Honor of Wilfried Kurz, eds. M. Rappaz, C. Beckermann, and R. Trivedi. Warrendale, PA : TMS, 2004. p. 373-378. , Feb-2004